

Lecture Notes

EC226 Term 2 2024/25

Panel Data Analysis

Reading:

- Stock and Watson (2003): Chapter 10
- Dougherty (2016): Chapter 14
- Wooldridge (2013): Chapter 13 + 14

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1 Unobserved Heterogeneity

Up to this point we have talked about either cross-sectional (observations on multiple individuals at a single point in time) or time series (data for a single unit of observation at multiple points of time). Panel data combines these elements together so that we have observations on multiple individuals, i , at multiple points of time, t .

Suppose we collect data from the labour force survey from 2010 and 2012, in which case we have observations on multiple individuals at two points in time. However, this is NOT panel data, but rather pooled cross-section data as the individuals are drawn independently of one another and we treat this just like cross-sectional data, although one may include a dummy for the year the individuals are drawn as another explanatory variable.

An example of a panel dataset can be seen in Figure 1, where for each individual, denoted id , we have 2 time observations (denoted $year$) one for 1975 and one for 1981 on a variety of variables including: the number of years in education ($educ$), whether they are in good health ($dghlth$), gender ($male$), marital status ($married$), whether they have young children ($ynghid$), number of hours of sleep per week ($sleep$), number of hours work per week ($work$), etc. The data is used for looking at the relationship between sleep and work patterns.

Figure 1: Sample panel dataset

	id	year	educ	dghlth	male	married	ynghid	sleep	work	sleep_day	work_day	d
1	1	75	16	1	1	0	0	66.51667	24.16667	3.192381	6.823333	
2	1	81	16	1	1	1	0	78.25	40.5	11.17857	8.1	
3	2	75	16	1	1	1	0	27.28224	46.21667	1.348477	9.043333	
4	2	81	16	1	1	0	0	26.58333	42.5	1.22619	8.7	
5	3	75	15	1	1	1	0	54.75	41.55	7.821429	8.309999	
6	3	81	15	0	1	1	0	51.91667	0	7.416667	0	
7	4	75	16	1	1	1	0	52.62334	46.3	7.119948	9.26	
8	4	81	16	1	1	1	0	56.45	29.78333	8.064286	5.956666	
9	5	75	17	1	1	1	0	62.28224	51.96667	8.911905	10.19233	
10	5	81	17	1	1	1	0	62.23333	0.2	9.047619	1.84	
11	6	75	16	1	1	1	1	61.33333	39.13334	8.761905	7.826667	
12	6	81	16	1	1	1	0	55.23333	42.78333	7.904762	8.556666	
13	7	75	16	1	1	1	1	42.96667	26.18333	8.991158	5.216667	
14	7	81	16	1	1	1	0	51.11666	37.86666	7.302381	7.573333	
15	8	75	16	0	1	0	0	64.5	67.01667	8.071428	11.40333	
16	8	81	16	1	1	0	0	65.16666	6.666667	9.109124	1.111333	
17	9	75	17	1	0	1	1	54.96667	30.88333	7.852381	6.176667	
18	9	81	19	1	0	1	0	52.66666	42.61666	7.512381	8.712333	
19	10	75	14	1	0	1	0	66.01667	0	8.17381	0	
20	10	81	14	1	0	1	0	60.56667	21.86667	8.652381	4.373333	
21	11	75	16	1	1	1	0	46.45	46	6.664286	9	
22	11	81	16	1	1	1	1	57.5	54.86666	8.214286	10.97333	
23	12	75	17	1	0	1	0	52.96667	42.1	7.709524	8.42	
24	12	81	19	1	0	1	0	55.63333	24.88333	7.861905	6.916667	
25	13	75	12	1	1	1	0	52	49.13334	7.428571	9.826667	
26	13	81	12	1	1	1	0	54.5	53.45	7.785714	10.69	
27	14	75	16	1	1	0	0	72.66666	42.83333	10.78095	8.566667	
28	14	81	16	0	1	0	0	66.83334	45.08333	9.54762	9.016666	

Within this dataset we could analyse (consistent with Term 1):

1. the cross-section of observations on all individuals for 1975
2. the cross-section of observations on all individuals for 1981
3. the “average” cross-section of observations for all individuals, obtained as the time average across 1975 and 1981

4. the cross-section of observations for all individuals based on the difference across the two time periods 1981 and 1975

From the data we have summary statistics on the variables sleep and work based on the 4 types of data discussed above would be:

```
. bys year: su sleep work
```

```
-----
```

```
-> year = 75
```

Variable	Obs	Mean	Std. Dev.	Min	Max
sleep	239	56.16109	8.380609	34.21667	101.8333
work	239	36.40342	15.3772	0	80.08334

```
-----
```

```
-> year = 81
```

Variable	Obs	Mean	Std. Dev.	Min	Max
sleep	239	55.28347	9.082267	20.66667	95.45
work	239	31.71855	16.68944	0	69.91666

```
. su sleep work
```

Variable	Obs	Mean	Std. Dev.	Min	Max
sleep	239	55.72228	6.907997	36.98333	85.00833
work	239	34.06098	13.2439	0	71.49167

```
. su dsleep dwork
```

Variable	Obs	Mean	Std. Dev.	Min	Max
dsleep	239	-.8776156	10.7034	-40.56667	32.33333
dwork	239	-4.684868	18.12148	-72.08334	46.61666

We can also look at the regression of the relationship between sleep and work (across the 4 data types noted above) using OLS:

$$sleep_{it} = \beta_0^t + \beta_1^t work_{it} + \varepsilon_{it} \quad t = 1975, 1981 \quad i = 1, \dots, N$$

.bys year: reg sleep work

-> year = 75						
Source	SS	df	MS	Number of obs = 239		
-----+-----				F(1, 237) = 33.91		
Model	2092.58727	1	2092.58727	Prob > F = 0.0000		
Residual	14623.2503	237	61.7014779	R-squared = 0.1252		
-----+-----				Adj R-squared = 0.1215		
Total	16715.8375	238	70.2346115	Root MSE = 7.855		

sleep	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----						
work	-.1928306	.0331117	-5.82	0.000	-.2580616	-.1275997
_cons	63.18078	1.308093	48.30	0.000	60.60381	65.75776

-> year = 81

Source	SS	df	MS			
Model	999.00162	1	999.00162	Number of obs = 239		
Residual	18633.042	237	78.6204305	F(1, 237) = 12.71		
				Prob > F = 0.0004		
				R-squared = 0.0509		
				Adj R-squared = 0.0469		
Total	19632.0437	238	82.4875784	Root MSE = 8.8668		

sleep	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
work	-.1227588	.034438	-3.56	0.000	-.1906024	-.0549152
_cons	59.1772	1.233744	47.97	0.000	56.7467	61.60771

Or using time averaged values:

$$\overline{sleep}_i = \beta_0 + \beta_1 \overline{work}_i + \varepsilon_i \quad i = 1, \dots, N$$

reg sleep work						
Source	SS	df	MS	Number of obs = 239		
Model	654.046853	1	654.046853	F(1, 237) = 14.48		
Residual	10703.4137	237	45.162083	Prob > F = 0.0002		
Total	11357.4605	238	47.7204223	R-squared = 0.0576		
				Adj R-squared = 0.0536		
				Root MSE = 6.7203		
sleep	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
work	-.1251699	.0328914	-3.81	0.000	-.1899668	-.0603731
_cons	59.98569	1.201693	49.92	0.000	57.61833	62.35306

Or using differenced values over time:

$$\Delta sleep_i = \beta_1 \Delta work_i + \Delta \varepsilon_i \quad i = 1, \dots, N$$

reg d.sleep d.work

. reg d.sleep d.work						
Source	SS	df	MS	Number of obs = 239		
Model	3072.72222	1	3072.72222	F(1, 237)	=	30.02
Residual	24258.9275	237	102.358344	Prob > F	=	0.0000
Total	27331.6497	238	114.838864	R-squared	=	0.1124
				Adj R-squared	=	0.1087
				Root MSE	=	10.117

D.sleep	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
work						
D1.	-.1920311	.0350487	-5.48	0.000	-.2610778	-.1229844
_cons	-.5954273	.6547275	-0.91	0.364	-1.885256	.6944015

Extending the range of alternative ways of looking at the data, we could pool all observations together:

$$sleep_{it} = \beta_0 + \beta_1 work_{it} + \varepsilon_{it} \quad i = 1, \dots, N \quad t = 1975, 1982$$

reg sleep work

reg sleep work						
Source	SS	df	MS	Number of obs = 478		
Model	2733.37353	1	2733.37353	F(1, 476)	=	38.60
Residual	33706.5476	476	70.8120749	Prob > F	=	0.0000
Total	36439.9212	477	76.3939647	R-squared	=	0.0750
				Adj R-squared	=	0.0731
				Root MSE	=	8.415

sleep	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
work	-.1477616	.023783	-6.21	0.000	-.1944942	-.101029
_cons	60.75519	.8968595	67.74	0.000	58.99289	62.51748

But which of these data sources is best?

Assuming the usual assumptions are correct the last model is best as it uses all observations and estimates the fewest parameters and therefore will be the most efficient. However, in reporting the final model we are assuming:

1. the parameters are constant across the two period (structural change?)
2. $E(\varepsilon_{it}|x_{it}) = 0$ is exogeneity of the explanatory variables.

Consider the general panel data model:

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it} \quad (1)$$

where y_{it} $i = 1, \dots, N$; $t = 1, \dots, T$ is the dependent variable and x_{it} is an explanatory variable. We can generalise this model to allow dummies for the different individuals (i.e. N-1) as well as for the different time periods, (i.e. T-1). In which case we have:

$$y_{it} = \alpha + \beta x_{it} + a_i + d_t + \varepsilon_{it} \quad (2)$$

where $d_t = \sum_{h=1}^{T-1} \delta_h d_{ht}$ and $a_i = \sum_{j=1}^{N-1} \delta_j a_{ji}$.

The term d_t (is constant over individuals, but varies over time) is included as average behaviour may well change over time (this is unobserved time heterogeneity) and could account for e.g. averaged reduced sleep in 1981 due to a particularly warm summer which meant people went to bed later to allow bedroom temperature to fall.

The term a_i (is constant over time but varies over individuals) is included as individuals are fundamentally different (this is unobserved individual heterogeneity and could account for why an individual is organised. As more organised people are likely to work more $cov(work_{it}, a_i) > 0$, but organised people are likely to ensure they are in bed at the right time and get the required amount of sleep, $cov(sleep_{it}, a_i) > 0$.

Imagine now a situation in which you estimate the the following (F) equation, given the true (T) equation:

$$y_{it} = \alpha + \beta x_{it} + v_{it} \quad (F)$$

$$y_{it} = \delta_0 + \delta_1 x_{it} + a_i + \varepsilon_{it} \quad (T)$$

We can think of the (T) equation being like:

$$y_{it} = \delta_0 + \delta_1 x_{it} + \delta_2 z_{it} + \varepsilon_{it}$$

In which case this is a standard omitted relevant variable problem and we know from Term 1:

$$b_1 = \delta_1 + \delta_2 \frac{cov(x_{it}, z_{it})}{V(x_{it})}$$

and so $\delta_2 \frac{cov(x_{it}, z_{it})}{V(x_{it})}$ is the BIAS, where δ_2 is the impact positive changes in a_i are likely to have on y_{it} (in the sleep example $\delta_2 > 0$, and we get an upward biased estimator).

Throughout the module thinking about potential sources of bias caused by correlation between the explanatory variable and the error term has been critical and we have seen a number of instances where this arises:

1. Omitted relevant variables (handout 5, Term 1)
2. Heterogeneity bias (handout 6, Term 2)

3. Errors in variables (handout 8, Term 1)
4. Endogeneity bias (handout 9: Term 1)
5. Sample selection bias (handout 6, Term 2)

2 Panel Data Estimators

Let us consider different methods for estimating the panel data model equation (2) (ignoring the time heterogeneity element).

2.1 Same intercept and same slopes (Pooled OLS)

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it} \quad (3)$$

Estimate by OLS using all NT observations. This is the pooled model and the OLS parameter estimates are consistent for large N , or large T or large NT .

$$b = \frac{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - x_{..})(y_{it} - y_{..})}{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - x_{..})^2} \quad (4)$$

$$a = y_{..} - bx_{..}$$

where

$$x_{..} = \frac{\sum_{i=1}^N \sum_{t=1}^T x_{it}}{NT}, \quad y_{..} = \frac{\sum_{i=1}^N \sum_{t=1}^T y_{it}}{NT}$$

(see Appendix A for an example of estimating a panel data using pooled OLS and Appendix B (C) contains some Stata (R) code for panel data methods).

2.2 Different intercepts and different slopes

$$y_{it} = \alpha_i + \beta_i x_{it} + \varepsilon_{it} \quad (5)$$

This is not a particularly interesting case and can only be estimated if T is large enough to carry out N separate OLS regressions for each cross-sectional unit. If N is very large this will clearly be very tedious and impractical. This regression will also provide separate estimates of the error variance (one for each i). We have the standard OLS 2-variable regression formula for the coefficients estimates based on T observations:

$$b_i = \frac{\sum_{t=1}^T (x_{it} - x_{i.})(y_{it} - y_{i.})}{\sum_{t=1}^T (x_{it} - x_{i.})^2} \quad (6)$$

$$a_i = y_{i.} - b_i x_{i.}$$

where

$$x_{i.} = \frac{\sum_{t=1}^T x_{it}}{T}, \quad y_{i.} = \frac{\sum_{t=1}^T y_{it}}{T}$$

NOTE:

1. $\hat{\beta}_i$ and $\hat{\alpha}_i$ are consistent only for large T .
2. This model can be run by estimating N separate OLS equations or by running one large equation using additive as well as interactive dummies (defined for each individual, i). The coefficient estimates will be exactly the same as before, but in this case you get one estimated error variance.

2.3 Different intercepts and same slopes (One-Way Fixed effects (FE))

$$y_{it} = \alpha + \beta x_{it} + a_i + \varepsilon_{it} \quad (7)$$

To estimate this model we create $N - 1$ dummies for the N individuals and regress y on an intercept, the $N - 1$ dummies and x :

$$b = \frac{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - x_{i.})(y_{it} - y_{i.})}{\sum_{i=1}^N \sum_{t=1}^T (x_{it} - x_{i.})^2} \quad (8)$$

$$a_i = y_{i.} - bx_{i.}$$

where:

$$x_{i.} = \frac{\sum_{t=1}^T x_{it}}{T}, \quad y_{i.} = \frac{\sum_{t=1}^T y_{it}}{T}$$

and the slope coefficients can be estimated consistently by OLS, providing N is large or T is large or NT is large, however the intercepts are only estimated consistently if T is large. The model could alternatively be estimated using the idea of partitioned regression, by following the three steps:

1. Run a regression of y_{it} on a_i and taking the residuals (which would be $y_{it} - y_{i.}$, i.e. individual deviations from mean for y)
2. Run a regression of x_{it} on a_i and taking the residuals (which would be $x_{it} - x_{i.}$, i.e. individual deviations from mean for x)
3. Estimate:

$$y_{it} - y_{i.} = \beta(x_{it} - x_{i.}) + (\varepsilon_{it} - \varepsilon_{i.}) \quad (9)$$

and this gives exactly the coefficient estimates in equation (8) and is known as the Within-Group estimator and we denote this estimator as b^w .

(see Appendix A for an example of estimating a panel data using FE).

An equivalent method to obtain the FE estimator (see Mundlak (1979)) is to estimate the model:

$$y_{it} = \alpha + \beta x_{it} + \gamma x_{i.} + \varepsilon_{it} \quad (10)$$

where as noted above:

$x_{i.} = \frac{\sum_{t=1}^T x_{it}}{T}$ this works because of partitioned regression as a regression of x_{it} on $x_{i.}$ gives $x_{it} - x_{i.}$

NOTE:

You cannot include explanatory variables which are time invariant with your fixed effects model as these variables would be perfectly correlated with the fixed effect terms.

2.4 One-Way Random Effects (RE) model

$$y_{it} = \alpha + \beta x_{it} + a_i + \varepsilon_{it} \quad (11)$$

where a_i 's are random drawings from some distribution, and we are required to say something about this distribution of a_i :

1. $E(a_i) = 0$
2. $V(a_i) = E(a_i^2) = \sigma_a^2$
3. $V(v_{it}) = V(a_i + \varepsilon_{it}) = \sigma_a^2 + \sigma^2$
4. $cov(v_{it}, v_{is}) = cov(a_i + \varepsilon_{it}, a_i + \varepsilon_{is}) = V(a_i) = \sigma_a^2$ as $(cov(a_i, \varepsilon_{is}) = 0$

The RE estimator exploits the non-zero covariance in the error terms, i.e. $cov(v_{it}, v_{is})$ to transform the equation (11), such that the error term is serially uncorrelated within i . Define:

$$\lambda = 1 - \sqrt{\frac{\sigma^2}{\sigma^2 + T\sigma_a^2}}$$

(which is clearly related to the $corr(v_{it}, v_{is}) = \frac{\sigma_a^2}{\sigma^2 + \sigma_a^2}$)

Then the RE estimator uses the following transformed version of equation (11):

$$y_{it} - \lambda y_{i.} = (1 - \lambda)\alpha + \beta(x_{it} - \lambda x_{i.}) + (\eta_{it} - \lambda \eta_{i.}) \quad (12)$$

If

1. $\lambda = 0$ (which implies $\sigma_a^2 = 0$) then equation (12) becomes:
 $y_{it} = \alpha + \beta x_{it} + \eta_{it}$ (and this is POOLED OLS)
2. $\lambda = 1$ (which implies $T\sigma_a^2 \rightarrow \infty$) then equation (12) becomes:
 $y_{it} - y_{i.} = \beta(x_{it} - x_{i.}) + (\eta_{it} - \eta_{i.})$ (which is the WITHIN GROUP i.e. FE estimator).
3. $0 < \lambda < 1$ RE is the most efficient estimator.

(see Appendix A for an example of estimating a panel data using RE).

2.5 First difference model

Consider the model:

$$y_{it} = \alpha + \beta x_{it} + a_i + \varepsilon_{it} \quad i = 1, \dots, N \quad t = 1, \dots, T$$

then lagging this equation by 1, we get:

$$y_{it-1} = \alpha + \beta x_{it-1} + a_i + \varepsilon_{it-1} \quad i = 1, \dots, N \quad t = 2, \dots, T$$

subtracting these two equations from each other, we get:

$$\Delta y_{it} = \beta \Delta x_{it} + \Delta \varepsilon_{it} \quad i = 1, \dots, N; \quad T = 2, \dots, T \quad (13)$$

this has eliminated the unobserved heterogeneity (a_i) and equation (13) can be estimated by OLS:

$$b = \frac{\sum_{i=1}^N \sum_{t=2}^T (\Delta x_{it} - \Delta x_{..})(\Delta y_{it} - \Delta y_{..})}{\sum_{i=1}^N \sum_{t=2}^T (\Delta x_{it} - \Delta x_{..})^2} \quad (14)$$

where:

$$\Delta x_{..} = \frac{\sum_{i=1}^N \sum_{t=2}^T \Delta x_{it}}{n(T-1)}, \quad \Delta y_{..} = \frac{\sum_{i=1}^N \sum_{t=2}^T \Delta y_{it}}{n(T-1)}$$

NOTE:

For $T = 2$ (without an intercept in the model) this will exactly reproduce the Fixed Effects Estimator, however, for $T > 2$ the two will be different. In particular for this model you have to worry about the MA(1) error process, $\Delta \varepsilon_{it} = \varepsilon_{it} - \varepsilon_{it-1}$. (see Appendix A for an example of undertaking a first difference regression).

2.6 Two-Way Fixed Effects (TWFE)

In this case we have a model:

$$y_{it} = \alpha + \beta x_{it} + a_i + d_t + \varepsilon_{it} \quad (15)$$

where a_i are the unit-specific effects (individual heterogeneity) and d_t are the time-specific effects (time heterogeneity). To estimate this model we create $N - 1$ dummies for the N individuals and $T - 1$ dummies for the T time periods and regress y on an intercept, the $N - 1$ individual dummies, the $T - 1$ time dummies and x . In this case the OLS estimator

(TWFE estimator) is:

$$b = \frac{\sum_{i=1}^N \sum_{t=1}^T \tilde{x}_{it} \tilde{y}_{it}}{\sum_{i=1}^N \sum_{t=1}^T \tilde{x}_{it}^2} \quad (16)$$

where $\tilde{x}_{it} = x_{it} - x_{i.} - x_{.t} + x_{..}$ and $\tilde{y}_{it} = y_{it} - y_{i.} - y_{.t} + y_{..}$ and

$$x_{i.} = \frac{\sum_{t=1}^T x_{it}}{T}, y_{i.} = \frac{\sum_{t=1}^T y_{it}}{T}, x_{.t} = \frac{\sum_{i=1}^N x_{it}}{N}, y_{.t} = \frac{\sum_{i=1}^N y_{it}}{N}, x_{..} = \frac{\sum_{i=1}^N \sum_{t=1}^T x_{it}}{NT}, y_{..} = \frac{\sum_{i=1}^N \sum_{t=1}^T y_{it}}{NT}$$

and this requires $N \rightarrow \infty$ and $T \rightarrow \infty$ in order to get consistent estimates of both the individual and time effects.

The model could alternatively be estimated using the idea of partitioned regression, by following the three steps:

1. Run a regression of y_{it} on a_i and d_t and taking the residuals (which would be $y_{it} - y_{i.} - y_{.t} + y_{..}$, i.e. individual and time deviations from mean for y (with the overall mean added back in which is just a constant))
2. Run a regression of x_{it} on a_i and d_t and taking the residuals (which would be $x_{it} - x_{i.} - x_{.t} + x_{..}$, i.e. individual and time deviations from mean for x (with overall mean added back in, which is just a constant))
3. Estimate:

$$y_{it} - y_{i.} - y_{.t} + y_{..} = \beta(x_{it} - x_{i.} - x_{.t} + x_{..}) + (\varepsilon_{it} - \varepsilon_{i.} - \varepsilon_{.t} + \varepsilon_{..}) \quad (17)$$

and this gives exactly the coefficient estimates in equation (15).

An equivalent method to obtain the TWFE estimator (which is a generalisation of Mundlak (1979), see Wooldridge (2021)) is to estimate the model:

$$y_{it} = \alpha + \beta x_{it} + \gamma x_{i.} + \delta x_{.t} + \varepsilon_{it} \quad (18)$$

where as noted above:

$x_{i.} = \frac{\sum_{t=1}^T x_{it}}{T}$ and $x_{.t} = \frac{\sum_{i=1}^N x_{it}}{N}$ this works because of partitioned regression as a regression of x_{it} on $x_{i.}$ and $x_{.t}$, gives $x_{it} - x_{i.} - x_{.t} + x_{..}$.

3 Hausman Tests FE vs RE

The Hausman test is appropriate where you are comparing two estimators, where under H_0 both estimators are consistent, but one is efficient, whereas under H_1 the efficient estimator becomes inconsistent, but the inefficient estimators remains consistent. We have seen this before if we compared OLS and IV, whereby under the null hypothesis that the explanatory variables are exogenous both estimators are consistent and OLS is efficient, but under the alternative (endogeneity) OLS is inconsistent, but IV remains consistent.

The Hausman test can also be used to compare the FE and RE estimators, where under $H_0 : cov(a_i, x_{it}) = 0$, both RE and FE are consistent and RE is efficient.

Whereas under

$H_1 : cov(a_i, x_{it}) \neq 0$ RE is inconsistent, but FE is consistent.

The Hausman test looks at the difference in the coefficient estimator from the two estimation methods, specifically in the case of a single explanatory variable the test is written as:

$$H = \frac{(b^{FE} - b^{RE})^2}{V(b^{FE}) - V(b^{RE})} \sim \chi_1^2$$

Under H_0 we know $b^{FE} \approx b^{RE} \Rightarrow H \rightarrow 0$, whereas under H_1 we know $b^{FE} \neq b^{RE} \Rightarrow H \rightarrow \infty$.¹

(see Appendix A for an example of undertaking this test).

¹Note if your model has 4 slope coefficients then the test would be χ_4^2

4 Dynamic Panel Data Models

Consider the first-order dynamic model:

$$y_{it} = \alpha + \beta_1 x_{it} + \beta_2 x_{it-1} + \beta_3 y_{it-1} + a_i + \varepsilon_{it} \quad (19)$$

Substantial complications arise with estimating this model. In the fixed effects model this arises because if we view this as a within groups estimator model:

$$y_{it} - y_{i.} = \beta_1(x_{it} - x_{i.}) + \beta_2(x_{it-1} - x_{i.}) + \beta_3(y_{it-1} - y_{i.}) + (\varepsilon_{it} - \varepsilon_{i.})$$

But this will yield biased and inconsistent estimators as from (19) we know that $cov(y_{it-1}, \varepsilon_{it-1}) > 0$, but ε_{it-1} enters into the calculation of $\varepsilon_{i.}$ and as such $cov(y_{it-1} - y_{i.}, \varepsilon_{it-1} - \varepsilon_{i.}) \neq 0$. The magnitude of the bias can be notable e.g. for $T = 10$ and $\beta_3 = 0.5$ the bias was as big as 33.3% (with the sign of the bias being negative for $\beta_3 > 0$). However, as T goes to infinity (for fixed N) these estimator is consistent. In panel data we normally have small T and large N .

The solution to this estimation problem is to take first differences of equation (19):

$$y_{it} - y_{it-1} = \beta_1(x_{it} - x_{it-1}) + \beta_2(x_{it-1} - x_{it-2}) + \beta_3(y_{it-1} - y_{it-2}) + (\varepsilon_{it} - \varepsilon_{it-1}) \quad (20)$$

while the model is still complicated due to the correlation between the lagged dependent variable and the disturbance term. The error term is an MA(1) and by construction this must be related to $(y_{it-1} - y_{it-2})$. However, there is a simple Instrumental Variable (IV) estimator, given enough observations one could use either $(y_{it-2} - y_{it-3})$, or y_{it-2} or y_{it-3} separately, as instruments for $(y_{it-1} - y_{it-2})$. This is valid as the error process is an MA(1) and therefore unrelated to y_{it-2} and higher lagged terms.

Estimating a dynamic panel data model in Stata entails using the command `xtabond` and it takes the form:

```
xtabond y l(0/1).x, lags(1)
```

Appendices

A Panel Data Model Estimation

$$sleep_{it} = \alpha + \beta_1 work_{it} + \beta_2 educ_{it} + \beta_3 marr_{it} + \beta_4 yngkid_{it} + \beta_5 gdhlth_{it} + d_t + a_i + \varepsilon_{it}$$

Pooled OLS ($a_i = 0$):

```
. reg sleep work educ marr yngkid gdhlth d81
```

Source	SS	df	MS	Number of obs =	478
Model	3559.10361	6	593.183935	F(6, 471) =	8.50
Residual	32880.8176	471	69.810653	Prob > F =	0.0000
Total	36439.9212	477	76.3939647	R-squared =	0.0977
				Adj R-squared =	0.0862
				Root MSE =	8.3553

sleep	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
work	-.1500954	.0244002	-6.15	0.000	-.1980421 -.1021487
educ	-.3376091	.1333363	-2.53	0.012	-.5996168 -.0756015
marr	-.7032314	.8846449	-0.79	0.427	-2.44157 1.035108
yngkid	1.237632	1.288764	0.96	0.337	-1.294806 3.77007
gdhlth	.4839554	.9929816	0.49	0.626	-1.467267 2.435177
d81	-1.283828	.7994198	-1.61	0.109	-2.854699 .2870425
_cons	65.97926	2.029439	32.51	0.000	61.99139 69.96714

Fixed effects ($a_i \neq 0$):

```

. xtreg sleep work educ marr yngkid gdhlth d81, fe

Fixed-effects (within) regression              Number of obs   =        478
Group variable: id                          Number of groups  =        239

R-sq:  within = 0.1552                      Obs per group:   min =         2
        between = 0.0416                      avg             =        2.0
        overall = 0.0736                      max             =         2

                                           F(6,233)        =        7.13
corr(u_i, Xb) = -0.1843                     Prob > F         =       0.0000

-----+-----
      sleep |          Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-----+-----
      work |   -.2266694     .036054    -6.29   0.000   -.2977029   -.1556359
      educ |   -.000408     .8126563    -0.00   1.000   -1.601501   1.600685
      marr |   1.736899     1.547589     1.12   0.263   -1.312158   4.785956
      yngkid |  1.577757     1.460875     1.08   0.281   -1.300456   4.45597
      gdhlth |  1.459631     1.276652     1.14   0.254   -1.055627   3.974888
      d81 |  -1.543901     .7644317    -2.02   0.045   -3.049983   -.0378198
      _cons |  61.61064     11.07483     5.56   0.000   39.79104   83.43025
-----+-----
      sigma_u |  6.9425988
      sigma_e |  7.0542981
      rho     |  .49202022      (fraction of variance due to u_i)
-----+-----
F test that all u_i=0:         F(238, 233) =      1.81             Prob > F = 0.0000

. estimates store fe

```

Note:

This last command save the FE coefficient estimates and the variance-covariance matrix in fe

Random effects ($a_i \neq 0$):

```
. xtreg sleep work educ marr yngkid gdhlth d81, re
```

Random-effects GLS regression	Number of obs	=	478
Group variable: id	Number of groups	=	239
R-sq: within = 0.1462	Obs per group: min =		2
between = 0.0777	avg =		2.0
overall = 0.0970	max =		2
	Wald chi2(6)	=	55.85
corr(u_i, X) = 0 (assumed)	Prob > chi2	=	0.0000

	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
sleep	-.1662668	.024632	-6.75	0.000	-.2145446 -.1179889
work	-.3315232	.1489684	-2.23	0.026	-.6234959 -.0395504
educ	-.3321166	.9187011	-0.36	0.718	-2.132738 1.468504
marr	1.226987	1.214654	1.01	0.312	-1.153691 3.607665
yngkid	.7407084	.969707	0.76	0.445	-1.159882 2.641299
gdhlth	-1.312238	.6883493	-1.91	0.057	-2.661378 .0369015
d81	65.98503	2.22694	29.63	0.000	61.6203 70.34975
_cons					
sigma_u	4.4334097				
sigma_e	7.0542981				
rho	.28314058	(fraction of variance due to u_i)			

```
. estimates store re
```

Note:

This last command save the RE coefficient estimates and the variance-covariance matrix in re

1st differenced model:

$$\Delta sleep_{it} = \beta_1 \Delta work_{it} + \beta_2 \Delta educ_{it} + \beta_3 \Delta marr_{it} + \beta_4 \Delta yngkid_{it} + \beta_5 \Delta gdhlth_{it} + \Delta \varepsilon_{it}$$

```
. reg d.sleep d.work d.educ d.marr d.yngkid d.gdhlth
```

Source	SS	df	MS	Number of obs = 239		
Model	4076.30495	5	815.26099	F(5, 233) = 8.19		
Residual	23189.6147	233	99.5262434	Prob > F = 0.0000		
Total	27265.9197	238	114.562688	R-squared = 0.1495		
				Adj R-squared = 0.1313		
				Root MSE = 9.9763		

D.sleep	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
work						
D1.	-.2266694	.036054	-6.29	0.000	-.2977029	-.1556359
educ						
D1.	-.000408	.8126563	-0.00	1.000	-1.601501	1.600685
marr						
D1.	1.736899	1.547589	1.12	0.263	-1.312158	4.785956
yngkid						
D1.	1.577757	1.460875	1.08	0.281	-1.300456	4.45597
gdhlth						
D1.	1.459631	1.276652	1.14	0.254	-1.055627	3.974888
_cons	-1.543901	.7644317	-2.02	0.045	-3.049983	-.0378198

Note:

the coefficient estimates and SE of this model are exactly the same as that of the FE model as we only have T=2 time periods.

Hausman Test: FE vs RE

```
. hausman fe re
```

	---- Coefficients ----			
	(b) fe	(B) re	(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
work	-.2266694	-.1662668	-.0604027	.0263278
educ	-.000408	-.3315232	.3311152	.7988859
marr	1.736899	-.3321166	2.069016	1.2454
yngkid	1.577757	1.226987	.3507697	.8116481
gdhlth	1.459631	.7407084	.7189221	.8303669
d81	-1.543901	-1.312238	-.2316628	.332462

```

b = consistent under Ho and Ha; obtained from xtreg
B = inconsistent under Ha, efficient under Ho; obtained from xtreg

Test:  Ho:  difference in coefficients not systematic

      chi2(6) = (b-B)'[(V_b-V_B)^(-1)](b-B)
           =      8.61
Prob>chi2 =      0.1969
(V_b-V_B is not positive definite)

```

Note:

As the RE estimator is more efficient the difference between the FE and RE variance-covariance matrix ought to be positive definite. In this case it is not, but we still have a statistic which is well defined and we are unable to reject the null hypothesis that the RE estimator assumptions are valid.

B Stata Commands for Panel Data Models

Data can be either long form with 1 observation for each cross section and time period

state	year	unem	tax
1	1980	.052	.18
2	1980	.074	.20
3	1980	.065	.25
4	1980	.031	.15
1	1990	.081	.16
2	1990	.032	.20
3	1990	.045	.22
4	1990	.043	.10
1	2000	.043	.18
2	2000	.055	.22
3	2000	.062	.22
4	2000	.066	.15

or wide with one observation for each cross sectional unit:

state	unem1980	unem1990	unem2000	tax1980	tax1990	tax2000
1	.052	.081	.043	.18	.16	.18
2	.074	.032	.055	.20	.20	.22
3	.065	.045	.062	.25	.22	.22
4	.031	.043	.066	.15	.10	.15

One can move between these two types of data

```
/* Making the data into wide from long */
reshape wide unem tax, i(state) j(year)
/* Making the data into long from wide */
reshape long unem tax, i(state) j(year)
```

Other useful panel data commands are:

```
/* This constructs the average of the variable unem across counties within a state */
bys state: gen unemp = mean(unem)
/* this produces aggregate summary statistics for unem and tax across States so
would be just time series data */
collapse (mean) munem=unem (mean) mtax=tax, by(year)
/* egen are extended generate commands and in this case produces a new series which
takes unique values for each gender/ethincity combination */
egen group=group(race sex)
```

Suppose data consist of a panel of 50 states observed over time for the variables: wage (wage), experience (exp) and education (educ). There are 4 options for doing FIXED EFFECT models in STATA.

```
/* This is a within group estimator in which you demean each variable for the time
averaged value*/
bys state: egen avg_wage = mean(wage)
gen d_wage = wage { avg_wage
bys state: egen avg_exp = mean(exp)
gen d_exp = exp { avg_exp
bys state: egen avg_educ = mean(educ)
gen d_educ = educ { avg_educ
reg d_wage d_exp d_educ
/* Using xi option, in which you construct a dummy variable for each state and
these are reported in the output table */
xi: reg wage experience education i.state
/* Use the absorb command to run the same regression as above, without reporting
state effects */
areg wage experience education, absorb(state)
/* Using xtreg, but required declaring the panel dataset first and stores the estimates */
xtset state year
xtreg wage exp educ, fe
estimates store fe
/* Random effects is similar and stores the estimates */
xtreg wage exp educ, re
estimates store re
/* Hausman test */
hausman fe re
/* POLS */
reg wage exp educ
estimates store pols
/* Creating an tables of output from FE, RE and POLS
outreg2 [pols fe re] using panel, replace excel
```

C R Commands for Panel Data Models

The following is a complete list of libraries that have been loaded:

```
library(readxl)
library(tseries)
library(urca)
library(forecast)
library(ggplot2)
library(dplyr)
library(zoo)
library(xts)
library(ggfortify)
library(dyn)
library(lmtest)
library(car)
library(MASS)
library(reshape2)
library(data.table)
library(stargazer)
```

Data can be either long form with 1 observation for each cross section and time period

state	year	unem	tax
1	1980	.052	.18
2	1980	.074	.20
3	1980	.065	.25
4	1980	.031	.15
1	1990	.081	.16
2	1990	.032	.20
3	1990	.045	.22
4	1990	.043	.10
1	2000	.043	.18
2	2000	.055	.22
3	2000	.062	.22
4	2000	.066	.15

or wide with one observation for each cross sectional unit:

state	unem1980	unem1990	unem2000	tax1980	tax1990	tax2000
1	.052	.081	.043	.18	.16	.18
2	.074	.032	.055	.20	.20	.22
3	.065	.045	.062	.25	.22	.22
4	.031	.043	.066	.15	.10	.15

One can move between these two types of data

```
# Making the data into wide from long
panelexample <- read_excel("C:/Users/ecrbl/Downloads/panelexample.xlsx")
panel_wide <- dcast(setDT(panelexample), year ~ state, value.var=c("unem", "tax"))
# Making the data into long from wide
panel_long <- melt(setDT(panel_wide),
  measure = patterns("unem", "tax"),
  variable.name = 'state', value.name = c('unem', 'tax'))
```

Other useful panel data commands are:

```
# This constructs the average of the variable unem across counties within a state
panel_long <- panel_long %>%#
  group_by(state) %>%
  mutate(m_unem = mean(unem))
)

# this produces aggregate summary statistics for unem and tax across States so
# would be just time series data
panel_ts <- panel_long %>%#
  group_by(state) %>%
  summarise(munem = mean(unem),
    mtax = mean(tax))
)
```

Suppose data consist of a panel of 30 states observed over 10 years for the variables: wage (wage), experience (exp) and education (educ). There are 3 options for doing FIXED EFFECT models in R.

```
state_wage <- read_excel("C:/Downloads/state_wage.xlsx")
state_wage <- state_wage %>%
  mutate(State=as.factor(State))
ols_fe <- lm(wage ~ educ + exp + State, data = state_wage)
summary(ols_fe)
wages_demeaned <- with(state_wage, data.frame(
  d_wage = wage - ave(wage, State),
```



```
d_educ = educ - ave(educ, State),
d_exp = exp - ave(exp, State)))
ols_fe2 <- lm(d_wage ~ d_educ + d_exp -1, data = wages_demeaned)
summary(ols_fe2)
ols_fe3 <- plm(wage ~ educ + exp,
               data = state_wage,
               index = c("State", "Year"),
               model = "within")
# print summary using robust standard errors
ols_fe4 <- coeftest(ols_fe3, vcov. = vcovHC, type = "HC1")
summary(ols_fe4)
# Random effects is similar and stores the estimates
ols_re <- plm(wage ~ educ + exp,
              data = state_wage,
              index = c("State", "Year"),
              model = "random")
summary(ols_re)
# Hausman test
phtest(ols_fe3, ols_re)
# POLS
ols <- lm(wage ~ exp + educ, data=state_wage)
summary(ols)
# Creating an tables of output from FE, RE and POLS
stargazer(ols, ols_fe4, ols_re, type='text', digits = 2,
          title = 'Table 2: Panel Data Regressions',
          style = 'qje',
          column.labels = c('POLS', 'FE', 'RE'), dep.var.labels = 'wages')
```

References

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